

AMOEBAS AND RELATIVES

Two of the protocist clades, the **Amoebozoa** and the **Opisthokonta**, have evolved different ways of moving about and obtaining food.

Amoebas of the clade Amoebozoa can change shape by oozing projections, called pseudopods, of their single cells. They use these “false feet” to creep forward and hunt smaller organisms, enveloping victims within a sac of fluid and digesting them while still alive. Some amoebas are giant cells, and are just visible to the naked eye. A few are gut parasites, causing amoebic dysentery in humans. One group, the slime moulds, have a remarkable strategy for avoiding starvation. When prey runs low their cells are attracted to one another by a chemical distress signal, and

aggregate into a tiny creeping slug. This then sprouts upward a number of stalks, which burst to scatter spores. Each spore hatches into a new amoeba ready to hunt on new ground.

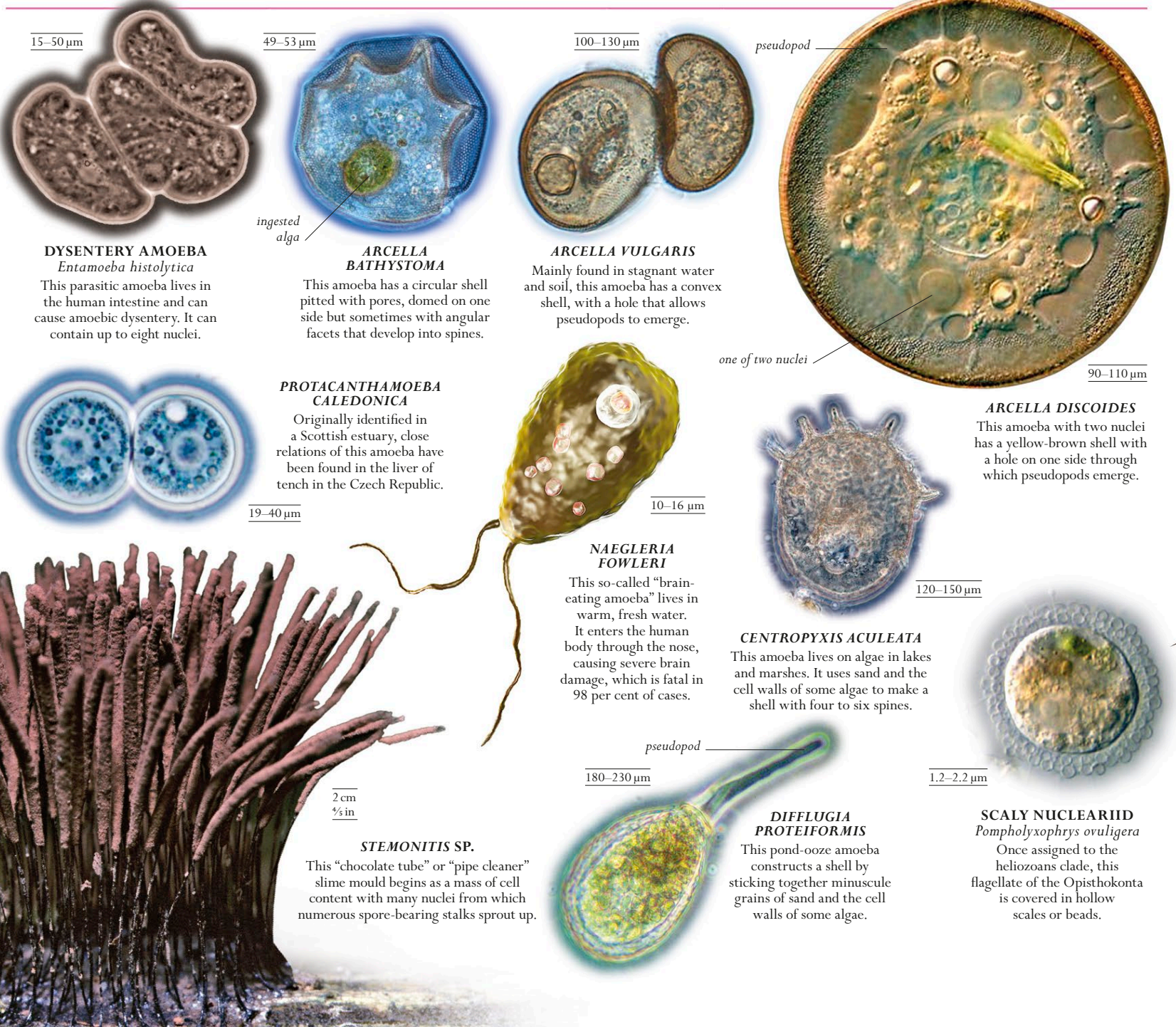
ANIMAL-FUNGUS ORIGINS

Most protocists of the clade Opisthokonta have developed a single whip-like flagellum for propulsion in open water. Early in the history of life, some of these protocists may have given rise to animals, and the single flagellum is still seen in the tail of animal sperm today. Others, called **nucleariids**, lost their flagellum and actually reverted to the amoeba-like state. The nucleariids may be closely related to fungi, as fungi also lack flagella and fertilize one another without the involvement of free-swimming sperm.

DOMAIN	EUKARYOTA
KINGDOM	PROTOCTISTA
CLADES	2
FAMILIES	About 50
SPECIES	About 4,000

DEBATE LIFE'S FIRST BRANCHES?

One theory of evolution says that the eukaryotes divided into two branches: unikonts (cells with one flagellum), such as opisthokonts, and bikonts (cells with two). Unikonts evolved into animals and fungi, while bikonts gave rise to plants. However, the DNA evidence for the theory is equivocal.



DYSENTERY AMOEBA
Entamoeba histolytica

This parasitic amoeba lives in the human intestine and can cause amoebic dysentery. It can contain up to eight nuclei.

ingested alga

ARCELLA BATHYSTOMA

This amoeba has a circular shell pitted with pores, domed on one side but sometimes with angular facets that develop into spines.

ARCELLA VULGARIS

Mainly found in stagnant water and soil, this amoeba has a convex shell, with a hole that allows pseudopods to emerge.

PROTACANTHAMOEBA CALEDONICA

Originally identified in a Scottish estuary, close relations of this amoeba have been found in the liver of tench in the Czech Republic.

NAEGLERIA FOWLERI

This so-called “brain-eating amoeba” lives in warm, fresh water. It enters the human body through the nose, causing severe brain damage, which is fatal in 98 per cent of cases.

CENTROPYXIS ACULEATA

This amoeba lives on algae in lakes and marshes. It uses sand and the cell walls of some algae to make a shell with four to six spines.

DIFFLUGIA PROTEIFORMIS

This pond-ooze amoeba constructs a shell by sticking together minuscule grains of sand and the cell walls of some algae.

ARCELLA DISCOIDES

This amoeba with two nuclei has a yellow-brown shell with a hole on one side through which pseudopods emerge.

SCALY NUCLEARIID
Pompholyxophrys ovuligera

Once assigned to the heliozoans clade, this flagellate of the Opisthokonta is covered in hollow scales or beads.

STEMONITIS SP.

This “chocolate tube” or “pipe cleaner” slime mould begins as a mass of cell content with many nuclei from which numerous spore-bearing stalks sprout up.